

**NEPVERSA: A Version-Aware Nepali Statutory Retrieval Benchmark Derived
from Amendment Footnotes**

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2026-09-19

Author Note

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The author thanks Prof. Achut Raj Kattel, Mahendra Morang Adarsh Multiple Campus, Tribhuvan University, for guidance on this research.

The author develops a commercial legal research tool for Nepali advocates. This research was carried out in the same code repository and reuses that project's Nepali language-processing components (text normalisation, stemming, romanisation, and Bikram Sambat date arithmetic). The corpus, extraction method, benchmark, and evaluation reported here are new work produced for this study. This relationship is disclosed as a competing interest.

Generative AI tools (Claude, Anthropic) were used for literature discovery, code assistance, and drafting; their use is described in the Method section. No benchmark item, gold answer, citation, or measurement was generated by a language model. The author reviewed and edited all content and takes full responsibility for it.

Data, extraction code, and benchmark annotations are available at <https://github.com/NikeGunn/nyasathi-reserch>. Because the licence status of the underlying statutory text is unresolved, the release comprises identifiers, offsets, annotations, and a rebuild script rather than the source text.

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Abstract

Retrieval-augmented generation systems assume that newer documents supersede older ones. Statutory law violates this assumption: when an Act is amended, the earlier text remains authoritative for events preceding the amendment. Recent benchmarks show that retrieval over a static consolidated corpus returns real but inapplicable provisions, and that detecting a non-governing version is substantially harder than selecting a governing one. These results come from high-resource languages with strong retrievers. Nepali has neither: it has no standard information retrieval benchmark, and published results show lexical retrieval outperforming multilingual dense retrieval on Nepali legal text. Constructing a version-aware Nepali corpus appears blocked, because commencement notices in the Nepal Gazette are published as scanned images and official Act PDFs extract to corrupted text through font-encoding damage. We show that the amendment record is nonetheless recoverable, from numbered footnotes carried inline in consolidated Acts, which name the amending instrument and its operation (insertion, substitution, or repeal) at clause granularity. We present **NEPVERSA**, the first version-aware Nepali statutory benchmark, whose gold answers are derived mechanically from these published statements and scored by deterministic nuggets rather than by a language-model judge. We report extraction accuracy against the published markup and release an automated cross-signal validator. Our central methodological finding is that a footnote parser that is approximately correct yields a corpus that appears entirely correct, because a missed amendment is indistinguishable from a provision never amended; we describe the redundant-signal design that makes such failures visible.

Keywords: legal information retrieval, low-resource languages, Nepali, temporal reasoning, benchmark construction, retrieval-augmented generation

Introduction

Retrieval-augmented generation (RAG) systems answer questions by retrieving documents and conditioning a language model on them (Lewis et al., 2020). Recent work has made retrieval adaptive: FLARE triggers retrieval when token probabilities fall (Jiang et al., 2023), Adaptive-RAG routes questions by predicted complexity (Jeong et al., 2024), IRCOT interleaves retrieval with chain-of-thought reasoning (Trivedi et al., 2023), and reinforcement-learned agents issue their own queries during reasoning (Jin et al., 2025; Yao et al., 2023).

These methods share an assumption that is invisible until it breaks: that a newer document supersedes an older one. In an encyclopedia this holds. In law it does not. When a legislature amends an Act, the earlier text remains the correct law for events that occurred before the amendment took effect. A question about a 2078 dispute must be answered from the 2078 text, and a system that retrieves today’s consolidated version returns a real provision that does not govern.

Recent work has begun to measure this failure. Cymbler et al. (2026) show that over a versioned corpus of the French tax code, static RAG retrieves the date-applicable version of an article 0% of the time while citing a real but inapplicable one. Sobhani et al. (2026) find that models resolve a version from an implicit date in 59.7% of cases but detect that a supplied version does *not* govern in only 26.7%. Prior et al. (2026) report the same pattern in German statutory question answering and conclude that temporal validity must be treated as a hard constraint rather than a ranking preference.

Every one of these results comes from a high-resource language with a strong retriever. Nepali is neither.

Why Nepali is the harder case

Nepali is spoken by over 26 million people and has no standard information retrieval benchmark at all (Acharya & Bal, 2026). What evidence exists suggests that the dense retrievers underpinning modern RAG perform poorly on it. Wagle et al. (2026), introducing the first RAG system for Nepali legal question answering, report **BM25 at P@1 0.91 against multilingual-E5-large at 0.75**, a lexical baseline beating a neural retriever by 16 points. Our own independent measurements on a different Nepali legal corpus reproduce the direction of that result (the Results section).

This matters for the temporal problem because every published mitigation assumes a retriever strong enough to surface candidate versions before a filter or an agent can choose between them. **If first-stage recall is weak, version selection never gets the chance to fail properly. It fails earlier, and for a different reason.** Whether the findings of Cymbler et al. (2026) and Sobhani et al. (2026) survive that regime is an open empirical question, and it cannot be asked without a version-aware corpus in such a language.

No such corpus exists. Existing Nepali legal NLP works over case law with no statutory versioning (Wagle et al., 2026) or provides an assistant evaluated on answer similarity with no temporal dimension at all (Thapa et al., 2026).

The obstacle, and the way around it

Building one is harder than it sounds, for a reason specific to how Nepal publishes law. A validity window is normally derived from gazette commencement notices. In Nepal those notices are overwhelmingly published as scanned images: of 53 gazette documents we examined, **50 contained no machine-readable text**. The consolidated Act PDFs on the Nepal Law Commission website are worse than unhelpful. All 61 extract to orthographically impossible text

(मिति → मममि) because the typesetting pipeline writes incorrect Unicode codepoints into real Unicode fonts. The documents look correct on screen and cannot be quoted.

Our contribution begins with the observation that the amendment record is already present in a different place: **inside the consolidated statutory text itself**, as numbered footnotes attached to individual clauses.

1(d1) Avail credit or advance facilities by the Chief Executive Officer ...

2(e) Re-avail or re-provide loans from or by another Bank ...

1 Added by the First Amendment.

2 Amended by the First Amendment.

Each marker names the amendment and, through its legend, the operation it performed (insertion, substitution, or repeal) **at clause granularity**. This is amendment provenance published as part of the law, and it can be extracted deterministically. Ground truth for a version-aware benchmark therefore does not require expert annotation of what changed: the statute states it.

Contributions

1. **A method for deriving provision versions from amendment footnotes** in consolidated statutes, with an explicit account of what the source supports and what it does not. Insertions and repeals yield fully determined validity windows; substitutions yield a determined *post*-state and an undetermined prior text, and we mark this rather than imputing it.
2. **NEPVERSA**, the first version-aware Nepali statutory benchmark, with gold answers derived mechanically from published amendment footnotes and scored by deterministic

nuggets rather than by an LLM judge, following Cymbler et al. (2026), who avoid a judge for temporal correctness on the grounds that it inherits the bias it is meant to measure.

3. **An extraction-accuracy evaluation** against the published HTML markup, together with a cross-signal validator. We report a finding we consider central for anyone building such a corpus: **a footnote parser that is approximately correct produces a corpus that appears entirely correct**, because a missed amendment is indistinguishable in the output from a provision that was never amended.
4. **A characterisation of the weak-retriever regime** in Nepali legal retrieval, with re-derived measurements, and an account of what it implies for the transferability of published version-aware RAG results.

All code, extraction rules, benchmark items and validation findings are released. Because the licence status of the underlying statutory text is unresolved (the Ethical Considerations section), we release identifiers, offsets, annotations and a rebuild script rather than the source text.

Related Work

Adaptive and agentic retrieval

A substantial line of work makes retrieval conditional. FLARE triggers retrieval when generated-token probabilities fall below a threshold (Jiang et al., 2023). Adaptive-RAG trains a classifier that routes each question to no retrieval, single-step retrieval, or multi-step retrieval according to predicted complexity (Jeong et al., 2024). IRCOT interleaves chain-of-thought reasoning with retrieval so that each reasoning step can issue a new query (Trivedi et al., 2023), and reinforcement-learned agents such as Search-R1 learn query policies directly (Jin et al.,

2025). ReAct established the general pattern of interleaving reasoning with tool calls (Yao et al., 2023).

These methods were developed and evaluated on English open-domain benchmarks with strong dense retrievers. The present work does not propose a new retrieval policy. It supplies the resource needed to ask whether the assumptions behind such policies hold in a low-resource, version-sensitive setting.

Temporal and version-aware legal retrieval

Three 2026 results establish that temporal validity is a distinct failure mode rather than a ranking nuisance.

Cymbler et al. (2026) construct a versioned corpus of 32,436 article-versions of the French tax code spanning 1938–2031 and show that RAG over a static current-version corpus retrieves the date-applicable version **0%** of the time while confidently citing a real but inapplicable one; an end-to-end retriever over a multi-version index reaches 98.3% strict accuracy, locating the problem in corpus construction rather than in model capability. They score answers against atomic ground-truth nuggets and explicitly decline to use an LLM judge, on the grounds that a judge would inherit the temporal bias it is meant to measure, a methodological choice we adopt.

Sobhani et al. (2026) introduce TIDE, 3,050 expert-verified question–answer pairs over 644 Bangladeshi customs instruments issued between 1969 and 2025, and name **version resolution** as the central challenge. Their sharpest result is asymmetric: models resolve a version from an implicit date in 59.7% of cases but recognise that a supplied version does *not* govern in only 26.7%. Detection of non-applicability is much harder than selection of the applicable version, which motivates our treatment of abstention as a first-class measured outcome.

Prior et al. (2026) study German statutory question answering and separate two failure modes (post-cutoff staleness and recency bias) concluding that reliable legal question answering requires treating temporal validity as a hard constraint. Fan et al. (2026) approach the same problem from the training side, learning temporal consistency through reinforcement over temporally-indexed data spanning multiple amendment periods.

Our work differs in jurisdiction, language, and in what it supplies. TIDE and FiscalQA Pro are benchmarks *plus* model evaluations in languages with mature retrieval infrastructure. We supply the missing resource for a language that has none, together with the extraction method that makes such a resource obtainable where the conventional source (gazette commencement notices) is unusable.

Nepali legal and low-resource NLP

Wagle et al. (2026) present the first RAG system for Nepali legal question answering, using case law from the Nepal Kanun Patrika archive. Two of their findings bear directly on this work. First, **BM25 reaches P@1 0.91 while multilingual-E5-large reaches 0.75**, a lexical method outperforming a neural one by a wide margin, which is the inverse of the English pattern the adaptive retrieval literature assumes. Second, their corpus is case law only: no statutes, no amendments, no versioning.

Thapa et al. (2026) introduce NepKANUN, a fine-tuned RAG assistant for Nepali legal queries evaluated by BERTScore (F1 0.82 / 0.77 / 0.71 across question difficulty). It is a system rather than a resource, and carries no temporal dimension.

Acharya & Bal (2026) argue that Nepali lacks any standardised information retrieval benchmark and propose building one covering monolingual, cross-lingual, and code-mixed

retrieval, noting that Nepali is a compelling testbed for morphology-aware retrieval. Their paper establishes the gap; it does not close it.

Comparable efforts in other low-resource legal settings, such as Urdu (Faisal & Yousaf, 2024), likewise build question-answering resources without a versioning dimension.

To our knowledge no prior work constructs a version-aware Nepali statutory corpus, and none uses Bikram Sambat validity windows. Systematic searches of arXiv for Nepali combined with amendment or statutory corpus terms, and for Bikram Sambat, returned no relevant prior work (the Related Work section).

Method

Corpus

Source selection

Nepali statutes are published in three forms, and only one is usable for a benchmark that requires verbatim quotation.

The Nepal Law Commission (lawcommission.gov.np) publishes consolidated Acts as PDFs. We attempted extraction from all 61 available Act PDFs using both pypdf and PyMuPDF. Every one produced orthographically impossible Nepali: मिति (“date”) extracts as मममि, कायन्वियन (“implementation”) as कायान्वयि. The cause is not scanning but font encoding. The typesetting pipeline maps legacy Nepali glyphs onto incorrect Unicode codepoints, lossily. The rendered pages are legible; the extracted text is not recoverable by any repair table we could construct, because the mapping is many-to-one.

The Nepal Gazette (राजपत्र) publishes commencement and amendment notices, which is where validity windows would normally come from. Of 53 gazette documents examined, 50

contained no machine-readable text layer, and the majority date from BS 2011–2013 (1950s CE) rather than the amendment period of interest.

We therefore harvest from **nepallaws.com**, which serves consolidated Acts as structured HTML with one page per provision and the citation hierarchy encoded in the URL path. As of our retrieval date (2026-09-19) the site carried **59 act entries (29 in Nepali and 30 in English)**, the two sets overlapping in content. Text extracted from this source is NFC Devanagari and passes character-exact quotation.

Amendment

provenance

Consolidated Nepali Acts carry amendment provenance inline, as superscript footnote markers attached to the unit each amendment touched, with a legend at the foot of the provision:

Amendment operation vocabulary, both published editions.

Nepali legend

English legend

Operation

संशोधनद्वारा थप।

Added by the ... Amendment.

INSERT

संशोधनद्वारा संशोधित।

Amended by the ... Amendment.

SUBSTITUTE

संशोधनद्वारा झिकिएको।

Taken out by the ... Amendment.

REPEAL

Two marker forms occur, both within a single Act:

Bracketed, for lettered clauses: ¹(d1)

Numbered, for sub-sections: ¹1)

A repealed unit leaves a positional trace (a run of elision dots where its text stood), so the corpus records not only that something was removed but where.

A second, independent signal comes from numbering. A section inserted by amendment carries a letter suffix (12A, 14B, १२क, १४ख) because it had to fit between existing numbers. Such a section did not exist before the amendment that created it, which is a validity window with a determined opening bound derived without reference to any footnote. We retain both signals and cross-check them (the Method, Validation section).

Version construction

Each amended unit expands into versions bounded by amendment events. Bounds are **half-open on amendment ordinals**, [from, to), so exactly one version governs any point; a closed upper bound would make both the pre- and post-amendment version match at precisely the boundary a point-in-time question is most likely to probe.

What the source

determines differs by
operation, and we encode the
difference rather than
smoothing over it:

What a consolidated edition determines about each period.

Operation

Prior period

Post period

INSERT

ABSENT (determined)

text printed (determined)

REPEAL

existed, **text not printed**

ABSENT (determined)

SUBSTITUTE

existed, **text not printed**

text printed (determined)

The consolidated edition prints only current text. For REPEAL and SUBSTITUTE the superseded wording is therefore **not in the document**, and we mark those periods PRESENT_TEXT_UNKNOWN. No benchmark item is generated that would require reproducing text we do not hold. This is enforced in code, not by convention: the item generator returns nothing for such a period.

Dates

A footnote names an amendment **ordinal** (“First Amendment”), not a date. Validity windows are therefore ordinal-valued by default. A Bikram Sambat or Gregorian date is attached only where the corresponding amending Act’s commencement provision has been independently verified; otherwise the date fields remain null. **No date is interpolated.** Bikram Sambat is a lunisolar calendar whose month lengths are tabulated rather than computed, so date arithmetic uses a tested conversion over the supported range BS 1975–2100.

Benchmark construction*Question types*

NEPVERSA defines five types. This release populates T2 and T3, which are the types derivable from amendment footnotes alone; T1, T4 and T5 are defined for completeness and discussed in the Limitations section.

T1 Lookup. Plain retrieval, no temporal component.

T2 Point-in-time. Which version governed at a given point.

T3 Supersession. Whether a unit still governs; the correct answer may be a refusal.

T4 Statute-to-case. From a provision to the judgments interpreting it.

T5 Cross-act. Requires combining provisions from more than one Act.

Gold derivation and scoring

Gold answers are **derived from published statements**, not authored. Where a footnote says a clause was added by the First Amendment, the gold answer for the prior period is that the clause was absent; the provenance string recording that footnote travels with the item.

Scoring uses deterministic **nuggets**: a closed label set, an exact citation key, or a verbatim span, following Cymbler et al. (2026), who avoid an LLM judge for temporal correctness

because a judge inherits the temporal bias it is intended to measure. Label and citation nuggets match case-insensitively; **exact nuggets do not**, since they exist to pin verbatim statutory wording.

Following Sobhani et al. (2026) we report **two scores that are never averaged**:

meaning-correct: the answer satisfies the content nuggets;

time-correct: the answer abstains where the governing version is absent.

Averaging them would conceal precisely the failure the benchmark exists to expose: a fluent answer quoting a real provision that did not govern at the queried time.

Status of items

Every generated item is status=unverified until audited by a human with legal training. We report the audited sample and its error rate; we do not claim expert verification of the full set.

Validation

Extraction correctness cannot be assessed by inspecting the output, because the output of a partially-correct parser is indistinguishable from that of a correct one: a missed amendment looks exactly like a provision that was never amended. We therefore validate in two ways.

Against markup. Markdown conversion flattens the superscript that carries the footnote marker, so `<sup>1</sup>6)` becomes 16), which is shape-indistinguishable from sub-section 16. We verify parser output against the published HTML, where `<sup>` delimits the marker explicitly.

Cross-signal. An automated validator compares three signals per provision (inline markers, parsed amendment units, and section numbering) and reports every disagreement, plus elision runs without a recorded repeal and footnotes attached to no unit. Findings are reported as

questions rather than failures, because some are genuine properties of the source (the Results section).

Reproducibility

Every provision record stores its source URL, retrieval timestamp, and the SHA-256 of its extracted text. Identity is the citation key `act:<slug>#<number>` rather than the URL: the site’s URL slugs changed during this study and the earlier forms now return 404, so a URL is not a stable identifier.

Harvesting is paced below the source API’s measured rate limit, and each run emits a report stating its own validity: a run with any transport failure is marked invalid rather than reported as a partial success. Raw responses are cached so that re-parsing after a parser change costs no further requests.

Results

All figures in this section are produced by `analysis/make_tables.py` from the released corpus and benchmark files and recorded in `paper/shared/numbers.json`. No value is entered by hand.

Corpus composition

Table 1 reports the corpus as counted from the harvested records.

Table 1

*orpus**Compos**ition by**Act*

Act

Provisions

Amended

%

Lettered

Repealed in full

INS

SUB

REP

Banking Offence and Punishment Act, 2064

34

6

17.6

5

1

6

13

2

Bonus Act, 2030

29

1

3.4

1

5

1

2

0

Foreign Exchange (Regulation) Act, 2019

46

3

6.5

21

1

1

1

2

Total**109****10****9.2****27**

7

8

16

4

Note. “Amended” counts provisions carrying at least one clause-level amendment footnote. “Lettered” counts sections whose number carries a letter suffix, indicating insertion by amendment. INS, SUB and REP are counts of individual amendment operations (insertion, substitution, repeal), of which there are 28 in total. Harvesting retrieved 34 of 34, 29 of 29, and 47 of 47 provisions respectively, with one provision rejected by the quotability gate and no transport failures.

Amendment style varies sharply between Acts

The two Acts differ in *how* they were amended, not merely how much.

The Banking Offence and Punishment Act, 2064 is amended predominantly at clause level: 21 of its 24 amendment operations modify individual clauses within surviving sections, and it carries five sections inserted by amendment (12A, 14A, 14B, 19A, 19B). Its amendment history is shallow, since every footnote in the harvested edition attributes to the First Amendment.

The Bonus Act, 2030 shows a second pattern. Only one provision carries clause-level footnotes, but **five of its 29 sections are repealed in their entirety**, and its footnotes attribute to amendments as late as the Fourth.

The Foreign Exchange (Regulation) Act, 2019 shows a third. It carries only 3 provisions with clause footnotes, but **21 of its 46 sections are amendment-inserted by numbering** (12A,

12B, and so on). At BS 2019 (1962 CE) it is the oldest Act in the corpus, and its amendment history is recorded almost entirely in section numbering rather than in footnotes.

The trend across the three is consistent: an older Act accumulates whole-section insertions and repeals, a newer one accumulates clause edits.

This has a practical consequence for corpus construction. An extraction pipeline tuned to clause-level footnotes recovers 21 operations from the Banking Act, 3 from the Bonus Act and 3 from the Foreign Exchange Act, which would suggest the two older Acts are barely amended. They are not: their amendments are recorded structurally, as a lettered section number or a section retained with its text replaced by an elision run, rather than in clause footnotes. **Both mechanisms must be handled, or an Act’s amendment history is systematically understated by the pipeline that measures it, and understated most for the oldest Acts, which have the longest history to record.**

Benchmark composition

Table 2

	<i>Benchmark</i>
	<i>Composition by</i>
	<i>Question Type</i>
Question type	
Items	
Abstention expected	

T2 point-in-time

36

12

T3 supersession

10

10

Total

46

22 (47.8%)

Note. Abstention-expected items are those for which the governing version is absent, so that the correct response is a refusal rather than a citation. Gold is scored by deterministic nuggets: 22 label nuggets, 24 citation nuggets and 24 verbatim-span nuggets across the set. All items carry status=unverified pending human audit.

Almost half of NEPVERSA items (47.8%) have abstention as the correct response. This is by construction rather than by design choice: an inserted clause did not exist before its inserting amendment, and a repealed clause does not exist after its repealing one, so a question anchored to those periods has no governing text to cite. It positions the benchmark directly on the behaviour Sobhani et al. (2026) found hardest: recognising that a supplied version does not govern, at 26.7% accuracy, against 59.7% for selecting one that does.

Extraction accuracy

Parser output was verified against the published HTML markup, in which <sup> tags delimit footnote markers explicitly. For Section 15 of the Banking Offence and Punishment Act, the most heavily amended provision in the corpus with six amended sub-sections, the parser

recovered sub-sections 1, 2, 3 and 6 as substitutions and 7 and 8 as insertions. This **matches the HTML markers exactly**, including the four cases that Markdown conversion renders ambiguous.

The cross-signal validator reports five findings over the released corpus, none of which is an extraction defect:

- **Four lettered sections without footnote provenance** (12A, 14A, 14B, 19A in the Banking Act; 12A in the Bonus Act). These sections are identifiable as amendment-inserted from their numbering, but the published pages carry no corresponding footnote. The insertion is real and its provenance is recorded only structurally. This is a property of the source, and is the reason the pipeline maintains numbering as an independent second signal.
- **Orphan footnotes.** Section 19 of the Banking Act and Section 1 of the Bonus Act each carry a footnote naming an amendment that attaches to no marked unit. Both were confirmed against the published page: the amendment is stated without identifying which unit it touched. The footnote is retained and no unit is invented.

Extraction defects

found during construction

Seven distinct defects were found and fixed while building this corpus. We report them because their distribution is itself a finding: **none produced a visible symptom in the output.**

#

Defect

How found

1

Repeal phrasing Taken out by the ... Amendment. absent from the vocabulary

Manual reading

2

Footnote marker with no separating space (1Amended by ...)

Manual reading

3

Wholly repealed section rejected by a minimum-length check

Manual reading

4

Unknown-verb guard triggered on ordinary prose about amendment

Adversarial test

5

Repealed-in-full flag never written to the output records

Cross-signal validator

6

Second inline marker form ($\langle \sup 1 \rangle$) unhandled

Cross-signal validator

7

Flat-structure Acts yielded zero provisions in discovery

Manual reading

Defects 5 and 6 were found by the automated validator after manual inspection had passed the same corpus; defect 5 was introduced by the fix for defect 3. Before defects 1–3 and 5–6 were corrected, the Banking Act yielded 2 amended provisions and 14 benchmark items with no T3 items at all; afterwards it yields 6 amended provisions and 30 items including 3 T3 items.

The uncorrected pipeline reported a complete, internally consistent corpus while recovering under a third of the published amendments.

Defect 7 is the most consequential for coverage. Acts on the source site follow two structures (provisions nested under chapters, and provisions listed directly) and a discovery routine written for the first returns *zero* provisions for the second, silently and indistinguishably from an Act with no sections.

Retrieval regime

We re-derived retrieval measurements on an independent Nepali legal corpus of 330 Supreme Court judgments (4,102 passages) using a lexical BM25 retriever with proximity scoring, over a 31-question labelled set with 2 unanswerable controls: **P@1 = 0.871, Recall@5 = 0.968, MRR = 0.925**, with both unanswerable questions correctly refused and no false refusals.

These figures are consistent in direction with Wagle et al. (2026), who report BM25 at P@1 0.91 against multilingual-E5-large at 0.75 on a different Nepali legal corpus. Two independent measurements now place a lexical baseline above a multilingual dense retriever on

Nepali legal text, the inverse of the pattern that adaptive-retrieval methods developed on English benchmarks assume.

We note this as context for interpreting version-aware results in this language, not as a controlled comparison: our measurement and Wagle et al.’s use different corpora and different labelled sets, and we did not run a dense retriever on NEPVERSA.

Discussion

A partially-correct parser produces an apparently-correct corpus

The result we consider most transferable is not a number but a failure mode.

Across the construction of this corpus, seven distinct extraction defects were found. Every one had the same observable signature: **none**. A provision whose amendment footnote was missed enters the corpus looking exactly like a provision that was never amended. There is no malformed field, no parse exception, no anomalous value, only a record that is quietly wrong about the one property the benchmark exists to measure.

This is a sharper problem than ordinary extraction noise. A missing *text* field is visible; a missing *amendment* is not, because absence of amendment is itself a valid and common state. The usual coverage check, “34 of 34 provisions parsed, zero rejected”, was true of the corpus at a point when it was recovering fewer than half the amendments actually published in it.

Three mitigations proved effective, in increasing order of value:

1. **Fail loudly on unrecognised vocabulary.** When a footnote names an amendment using a verb the parser does not know, raising is strictly better than skipping. A rejected provision is a visible gap; a skipped footnote is an invisible falsehood. This required narrowing the trigger to footnote *grammar* (“by the First Amendment”) rather than the

word “amendment”, since statutes contain ordinary prose about amendment that must not be rejected.

2. **Maintain redundant signals.** Amendment-inserted sections are identifiable two ways: from a footnote, and from a lettered section number (12A, 12^क). Neither is complete. Sections exist whose insertion is recorded only in the numbering, and footnotes exist that name no unit at all.
3. **Cross-check the signals automatically.** A validator comparing inline markers, parsed amendment units, and section numbering, reporting every disagreement, found defects that manual inspection had missed, including one in the code written to fix earlier defects. We report its findings as questions rather than failures, because a fraction are genuine properties of the source.

Format fidelity determines which facts survive

The published HTML marks an amended unit with a superscript: `<sup>1</sup>1`) If anyone commits.... Markdown conversion flattens the superscript into the surrounding text, so `<sup>1</sup>6`) becomes 16), shape-indistinguishable from sub-section 16.

The information remains *recoverable* from markdown, because un-prefixed siblings disambiguate the prefixed ones, and our parser does recover it correctly. But recovery by inference is not the same as reading a fact the source states outright, and the inference is brittle in ways that depend on which sub-sections happen to be amended.

The general lesson for corpus construction in this setting: **the serialisation you scrape decides which facts survive it.** A conversion that preserves prose perfectly can destroy exactly the markup that carries provenance, while leaving text that still reads correctly, and therefore passes every check that inspects text.

Implications for version-aware retrieval in low-resource settings

Cymbler et al. (2026) show that static RAG retrieves the date-applicable version 0% of the time over French tax law, and that a multi-version index largely solves it. Sobhani et al. (2026) show that recognising a non-governing version is far harder than selecting a governing one. Both results presume a retrieval stage that reliably surfaces candidates.

That premise is weaker in Nepali. Wagle et al. (2026) report BM25 outperforming multilingual-E5-large by 16 points on Nepali legal retrieval, and our independent measurements on a different Nepali legal corpus are consistent in direction. Where first-stage recall is the binding constraint, version-selection machinery may not be reached at all, which would make the *ordering* of interventions different from the high-resource case: corpus versioning and lexical retrieval quality before agentic version selection.

We state this as a hypothesis that NEPVERSA makes testable, not as a result. The present work supplies the resource; the comparison is future work (the Future Work section).

What the source cannot tell us

Consolidated editions print current text. For a substituted or repealed unit the superseded wording is therefore absent from the document, and we mark those periods as textually undetermined rather than imputing content.

This bounds the benchmark in a specific and, we would argue, honest way. Insertions and repeals yield fully determined validity windows and support questions of both kinds: *what governed* and *whether anything governed*. Substitutions support only the latter. A benchmark that asked models to reproduce superseded wording we never observed would score every system wrong for a reason attributable to the corpus rather than the system.

Similarly, footnotes name amendment *ordinals*, not dates. Validity windows are ordinal-valued unless a commencement date has been independently verified. We leave date fields null rather than interpolating, because an interpolated date propagates silently: every downstream check can be internally consistent and uniformly wrong about which year applies.

Limitations

Scale. This release covers a small number of Acts. The extraction method is not the constraint; retrieval quota is. Amendment density varies substantially across Acts, and the corpus is weighted toward Acts selected for expected density rather than sampled representatively. Aggregate amendment rates reported here should not be read as estimates of amendment rates in Nepali legislation generally.

Question-type coverage. Only T2 (point-in-time) and T3 (supersession) items are populated. These are the types derivable from amendment footnotes alone. T1 (lookup) requires no additional evidence and is straightforward to add; T4 (statute→case) requires linkage to judgment text; T5 (cross-act) requires multiple Acts in a single reasoning chain. Their definitions are included so that later releases extend rather than redefine the benchmark.

No date-anchored windows. Validity windows are ordinal-valued, bounded by “First Amendment” or “Second Amendment”, because the footnotes name ordinals and not dates. Attaching Bikram Sambat and Gregorian dates requires each amending Act’s commencement provision, which the gazette record does not supply in machine-readable form. Consequently the benchmark tests *version* resolution but not yet *date* resolution, which is the form the question takes in practice (“what was the law on 2078-03-15?”). This is the single largest gap between NEPVERSA and comparable resources such as Cymbler et al. (2026), whose corpus carries real date bounds.

Superseded text is unavailable. Consolidated editions print only current text. For substituted and repealed units the prior wording is not in the document, so those periods are marked textually undetermined and generate no quotation questions. A benchmark built from *amending Acts* rather than consolidated editions would recover this, at the cost of a much harder alignment problem.

No model evaluation. This paper reports a resource and an extraction evaluation. It does not evaluate retrieval or generation systems on NEPVERSA. The claims in the Results section concern extraction correctness, not model capability, and nothing here should be read as a finding about how models perform on Nepali version-aware retrieval.

Human verification is sample-based. Gold answers are derived mechanically from published footnotes rather than authored, so verification checks that the derivation is faithful rather than establishing the answer independently. We report an audited sample and its error rate; items are released with status=unverified where no human has checked them. Inter-annotator agreement is computed only over items labelled by two human annotators.

Single source. All statutory text comes from one aggregator (nepallaws.com). Its consolidations were not independently reconciled against the Nepal Law Commission’s own texts, because those are the PDFs whose extraction failure motivated the choice of source (the Method, Source Selection section). Errors in the aggregator’s consolidation would propagate into the corpus undetected by any check described here.

English and Nepali editions. The site publishes both. Amendment footnotes appear in both, and the extraction vocabulary covers both, but this release does not verify that the two editions agree provision-by-provision on which units were amended. Disagreement between them would be an informative signal and is left to future work.

Ethical considerations

Licence and redistribution. The licence status of Nepali statutory text is unresolved. Government legal texts are typically freely reproducible, but “typically” is not a licence, and no explicit terms are published by the source aggregator. We therefore release **identifiers, character offsets, amendment annotations, nugget gold, and a rebuild script**, not the statutory text itself. Anyone can reconstruct the corpus by running the harvester against the live source. Where licence terms are later clarified, a full release becomes possible without changing the benchmark.

Personal data. Statutory text contains no personal data. The judgment corpus referenced for future T4 items consists of published court reports; no additional personal information is introduced beyond what those reports already make public.

Intended use and misuse. NEPVERSA is a research resource for evaluating retrieval systems. It is not legal advice, and its items are not a substitute for professional judgment. A system scoring well on it has demonstrated version resolution on a specific corpus, not competence in Nepali law. We note the asymmetry deliberately: in legal practice a confident wrong citation is more damaging than an acknowledged gap, which is why abstention is measured as a first-class outcome rather than penalised as non-response.

Competing interests. The author develops a commercial legal research tool for Nepali advocates. This research was conducted in the same repository and reuses that project’s Nepali language-processing components (text normalisation, stemming, romanisation, and Bikram Sambat date arithmetic); the corpus, extraction method, benchmark and evaluation reported here are new work. The relationship is disclosed in full in the Author Note.

Use of generative AI tools

Generative AI tools were used for literature discovery, code assistance, and drafting (Anthropic, 2026). All extraction code was reviewed and tested by the author; every reported number is produced by a script in the released repository and recorded with its source and generating command. No benchmark item, gold answer, citation, or measurement in this paper was generated by a language model. Citations were verified against the arXiv API, with author lists taken from the API response rather than from model output; this process corrected two attribution errors present in earlier drafts. The author reviewed and edited all content and takes full responsibility for it.

Conclusion

Version-aware legal retrieval has been shown to fail in ways that ordinary retrieval evaluation cannot see: a system returns a real provision, correctly cited, that does not govern the period in question (Cymbler et al., 2026), and recognising non-applicability is markedly harder than selecting an applicable version (Sobhani et al., 2026). These results come from languages with mature retrieval infrastructure. Whether they transfer to a low-resource language whose strongest retriever is lexical has not been testable, because no version-aware corpus existed in such a language.

We have supplied one. NEPVERSA is built from a source of amendment provenance that is available where the conventional source is not: the numbered footnotes that consolidated Nepali Acts carry inline, which name the amending instrument and its operation at clause granularity. Gold answers are therefore *derived from published statements* rather than authored, which makes the resource constructible without large-scale expert annotation and makes every item auditable against a specific sentence in the law.

We are explicit about the resulting bounds. Insertions and repeals yield fully determined validity windows; substitutions determine the post-amendment text only, and we mark the prior period undetermined rather than impute it. Footnotes name amendment ordinals rather than dates, so windows are ordinal-valued until commencement dates can be independently verified. These are properties of what Nepal publishes, not of the method, and recording them honestly is what keeps the benchmark’s gold trustworthy.

The methodological finding we expect to transfer furthest concerns extraction itself. **A footnote parser that is approximately correct produces a corpus that appears entirely correct**, because a missed amendment is indistinguishable in the output from a provision that was never amended. Coverage statistics cannot detect this; only redundant signals, loud failure on unrecognised vocabulary, and automated cross-checking can. Any project deriving provenance from consolidated legal text, in any jurisdiction, inherits this problem, and inherits it silently.

Future work

Date anchoring. Attaching Bikram Sambat and Gregorian bounds to validity windows requires the commencement provision of each amending Act. Obtaining these in machine-readable form is the highest-value next step, and is primarily an access problem rather than a technical one.

Model evaluation. NEPVERSA makes testable the hypothesis raised in the Results section: that where first-stage recall is the binding constraint, version-selection machinery may not be reached, reordering the interventions that matter relative to the high-resource case. Evaluating single-pass and agentic systems at matched retrieval budgets is the natural continuation.

Scale and coverage. Extending to more Acts, populating T1, T4 and T5, and reconciling the Nepali and English editions provision-by-provision would strengthen the resource considerably. Edition disagreement in particular is a free validation signal that this release does not yet exploit.

Portability. The extraction method depends on consolidated text carrying inline amendment footnotes, which is a common convention well beyond Nepal. Testing it against another jurisdiction’s consolidations would establish whether the approach generalises or is an artefact of one publisher’s typesetting.

Acharya, P., & Bal, B. K. (2026). Towards building a standard benchmark for low-resource Nepali information retrieval. *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*.

<https://doi.org/10.1145/3805712.3808533>X

Anthropic. (2026). *Claude*. Large language model. <https://claude.ai>X

Cymbler, R., Guez, D., & Fabre, L. (2026). Temporal misgrounding in legal RAG: A versioned-corpus benchmark for french tax law. *arXiv Preprint arXiv:2608.09393*.

<https://doi.org/10.48550/arXiv.2608.09393>X

Faisal, F., & Yousaf, U. (2024). LEGAL-UQA: A low-resource Urdu-English dataset for legal question answering. *arXiv Preprint arXiv:2410.13013*.

<https://doi.org/10.48550/arXiv.2410.13013>X

Fan, W., Zhou, Y., Zhang, M., Weng, Y., Hu, Y., & Zheng, T. (2026). Can LLMs time travel? Enhancing temporal consistency in legal agentic search through reinforcement learning.

arXiv Preprint arXiv:2605.25920. <https://doi.org/10.48550/arXiv.2605.25920>X

Jeong, S., Baek, J., Cho, S., Hwang, S. J., & Park, J. C. (2024). Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics*. <https://doi.org/10.48550/arXiv.2403.14403>X

Jiang, Z., Xu, F. F., Gao, L., Sun, Z., Liu, Q., Dwivedi-Yu, J., Yang, Y., Callan, J., & Neubig, G. (2023). Active retrieval augmented generation. *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*. <https://doi.org/10.48550/arXiv.2305.06983>X

Jin, B., Zeng, H., Yue, Z., Yoon, J., Arik, S., Wang, D., Zamani, H., & Han, J. (2025). Search-R1: Training LLMs to reason and leverage search engines with reinforcement learning. *arXiv Preprint arXiv:2503.09516*. <https://doi.org/10.48550/arXiv.2503.09516>X

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33. <https://doi.org/10.48550/arXiv.2005.11401>X

Prior, M., Schultz, A., & Grabmair, M. (2026). Asking for an old friend: Diagnosing and mitigating temporal failure modes in LLM-based statutory question answering. *arXiv Preprint arXiv:2605.23497*. <https://doi.org/10.48550/arXiv.2605.23497>X

Sobhani, M. E., Sayeedi, Md. F. A., Chowdhury, F. H., Arefeen, M. A., Sadeque, F., & Bari, Md. F. (2026). Time present and time past: Benchmarking large language models on temporally evolving document understanding. *arXiv Preprint arXiv:2608.08512*. <https://doi.org/10.48550/arXiv.2608.08512>X

Thapa, B., Koirala, P., Raut, R., Regmi, S., & Bal, B. K. (2026). NepKANUN: A RAG-based nepali legal assistant. *arXiv Preprint arXiv:2609.15999*.

<https://doi.org/10.48550/arXiv.2609.15999>X

Trivedi, H., Balasubramanian, N., Khot, T., & Sabharwal, A. (2023). Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*.

<https://doi.org/10.48550/arXiv.2212.10509>X

Wagle, S., Adhikari, A., Khanal, R., Bhandari, B., Manandhar, P., Acharya, P., & Bal, B. K. (2026). Retrieval augmented generation framework for the nepali legal domain question answering. *arXiv Preprint arXiv:2606.07523*. <https://doi.org/10.48550/arXiv.2606.07523>X

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing reasoning and acting in language models. *International Conference on Learning Representations*. <https://doi.org/10.48550/arXiv.2210.03629>X